

Identifiers**

Identifiers

Identifiers in Python are names used to identify variables, functions, classes, modules, and other objects. An identifier is a sequence of one or more characters that may consist of letters (both uppercase and lowercase), digits (0-9), and underscores (_).

Rules for Naming Identifiers in Python :--

Start with a Letter or Underscore: An identifier must begin with a letter (a-z, A-Z) or an underscore (_). It cannot start with a digit.

```
# Example:--
```

```
xyz = 10
print(xyz)
```

```
# Example:--
```

```
Xyz = 20
print(Xyz)
```

```
# Example:--
```

```
_xyz = 30
print(_xyz)
```

```
# Output
```

```
10
20
30
```

```
# Example:--
```

```
1xyz = 10
print(xyz)
```

```
# output
```

```
ERROR!
```

```
Traceback (most recent call last):
```

```
  File "<main.py>", line 1
```

```
    1xyz = 10
```

```
    ^
```

```
SyntaxError: invalid decimal literal
```

```
# Example:--
```

```
$xyz = 30
print(_xyz)
```

output

ERROR!

Traceback (most recent call last):

File "<main.py>", line 4

```
    $xyz = 30
    ^
```

SyntaxError: invalid syntax

Subsequent Characters: The characters following the initial letter or underscore can be letters, digits, or underscores.

```
_xy123 = 10
print(_xy123)
```

Output:

10

```
_xy$123 = 10
print(_xy$123)
```

Output:

ERROR!

Traceback (most recent call last):

File "<main.py>", line 1

```
    _xy$123 = 10
    ^
```

SyntaxError: invalid syntax

Case Sensitivity: Identifiers in Python are case-sensitive. This means myVariable, MyVariable, and myvariable are considered three different identifiers.

Example:--

```
x = 10
print(x)
```

output

10

```
print(X)
```

output

ERROR!

Traceback (most recent call last):

```
File "<main.py>", line 4, in <module>
NameError: name 'X' is not defined
```

No Spaces or Special Characters: Identifiers cannot contain spaces or special characters like !, @, #, \$, %, etc., except for the underscore (_).

```
# Example:--
```

```
x_y =10
print(x_y)
```

```
# Output
```

```
10
```

```
# Example:--
```

```
x$y=10
print(x$y)
```

```
# Output
```

```
ERROR!
```

```
Traceback (most recent call last):
```

```
File "<main.py>", line 1
```

```
    x$y=10
    ^
```

```
SyntaxError: invalid syntax
```

No Keywords: Identifiers cannot be the same as Python keywords. Keywords are reserved words in Python that have predefined meanings, such as if, else, while, for, def, class, etc. You can check the list of keywords using the keyword module.

```
# Example:--
```

```
if = 10
print(if)
```

```
# Output
```

```
ERROR!
```

```
Traceback (most recent call last):
```

```
File "<main.py>", line 1
```

```
    if = 10
    ^
```

```
SyntaxError: invalid syntax
```

No Built-in Function Names: It is not advisable (though technically possible) to use the names of built-in Python functions and modules (like print, list, str, int, etc.) as identifiers, as this can lead to confusion and bugs.

Example 1:--

```
print = 10  
print(print)
```

Output

ERROR!

Traceback (most recent call last):

File "<main.py>", line 2, in <module>

TypeError: 'int' object is not callable